

UNLOCK DOOR USING KEYPAD SENSOR

An IOT Based Mini Project

Bachelor of Computer Applications (BCA)

(Under Mangalore University)

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Introduction

In today's world, security is a major concern in homes, offices, and other institutions. Traditional lock-and-key systems, while simple, are prone to being picked or keys being lost or copied. With the growing availability of smart devices and sensors, Internet of Things (IoT) has paved the way for smarter security solutions. One such solution is a **Keypad-based Smart Door Lock System**, which replaces the traditional lock with a digital interface to enhance convenience and safety.

Abstract

This project involves designing an **IoT-based door lock system** that uses a **keypad input** to allow or deny access to a door. The system is built using a **microcontroller such as Arduino**, a **4x4 matrix keypad**, and a **servo motor**. When the correct password is entered via the keypad, the microcontroller sends a signal to the locking mechanism to unlock the door. If the password is incorrect, the system will deny access.

Requirements Analysis

Hardware Requirements:

A) Arduino UNO



B) 4x4 Matrix Keypad



C) Servo motor



D) LCD Display



E) Jumper wires



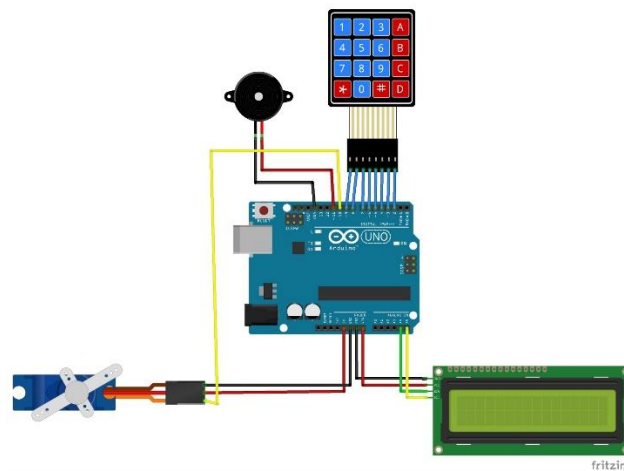
E) Power Supply

Software Requirements:

- Arduino IDE
- Embedded C
- Libraries: Keypad.h, Servo.h, Password .h ,o LiquidCrystal_I2C.h.

Implementation

- The keypad is interfaced with the microcontroller.
- A predefined password is stored in the microcontroller's memory.
- User enters a password using the keypad.
- If the input matches the stored password:
 - The microcontroller activates the servo to unlock the door.
 - Display "Access Granted" on the LCD.
- If the input is wrong:
 - Access is denied and a message like "Wrong Password" is displayed.
- After a short delay, the door locks again automatically.



CODE:

```
#include <Servo.h>
#include <LiquidCrystal_I2C.h>
#include <Keypad.h>
#include <Password.h>

#define BUZZER 11 // (unused, keep if you want a beep later)

Servo servo;
LiquidCrystal_I2C lcd(0x27, 16, 2);

// Password is 4 digits
Password password = Password("1111");
Const byte PASSWORD_LENGTH = 4;
Byte currentPasswordLength = 0;
```

```

Const byte startMaskCol = 5; // where stars begin

// Keypad setup
Const byte ROWS = 4;
Const byte COLS = 4;
Char keys[ROWS][COLS] = {
    {'D', 'C', 'B', 'A'},
    {'#', '9', '6', '3'},
    {'0', '8', '5', '2'},
    {'*', '7', '4', '1'},
};
Byte rowPins[ROWS] = {2, 3, 4, 5};
Byte colPins[COLS] = {6, 7, 8, 9};
Keypad keypad = Keypad(makeKeymap(keys), rowPins, colPins, ROWS,
COLS);

Void setup() {
    Servo.attach(10);
    Servo.write(50); // locked position
    Lcd.init();
    Lcd.backlight();
    Lcd.setCursor(3, 0);
    Lcd.print("WELCOME TO");
    Lcd.setCursor(0, 1);
    Lcd.print("DOOR LOCK SYSTEM");
    Delay(2000);
    Lcd.clear();
    showPrompt();
}

Void loop() {
    Char key = keypad.getKey();
    If (key == NO_KEY) return;

    Delay(60); // simple debounce

    If (key == 'C') {

```

```

    resetInput();
    return;
}

// Only process digits 0-9
If (key >= '0' && key <= '9') {
    // mask input
    Lcd.setCursor(startMaskCol + currentPasswordLength, 1);
    Lcd.print("*");

    Password.append(key);
    currentPasswordLength++;

    if (currentPasswordLength == PASSWORD_LENGTH) {
        // evaluate once
        If (password.evaluate()) {
            unlockDoor();
        } else {
            showError();
        }
        resetInput();
    }
}

Void showPrompt() {
    Lcd.setCursor(1, 0);
    Lcd.print("ENTER PASSWORD");
    Lcd.setCursor(startMaskCol, 1);
}

Void resetInput() {
    Password.reset();
    currentPasswordLength = 0;
    lcd.clear();
    showPrompt();
}

```

```

Void unlockDoor() {
  Servo.write(110); // unlocked
  Lcd.clear();
  Lcd.setCursor(0, 0);
  Lcd.print("CORRECT PASSWORD");
  Lcd.setCursor(0, 1);
  Lcd.print("DOOR UNLOCKED");
  Delay(5000);
  // optionally keep door unlocked or re-lock after delay:
  Servo.write(50); // auto re-lock
}

```

```

Void showError() {
  Lcd.clear();
  Lcd.setCursor(0, 0);
  Lcd.print("WRONG PASSWORD!");
  Lcd.setCursor(0, 1);
  Lcd.print("TRY AGAIN");
  Delay(2000);
}

```

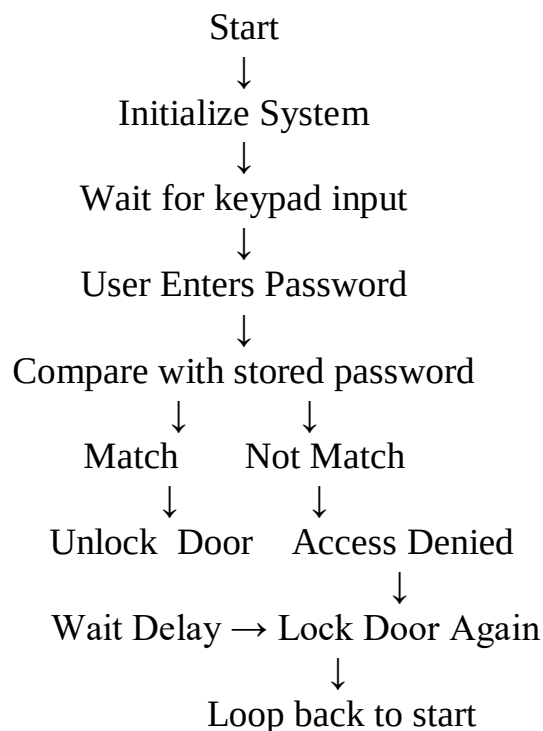
Code Explanation:

- **Libraries Used:**
 - Servo.h → Controls door lock servo motor.
 - LiquidCrystal_I2C.h → Displays messages on LCD.
 - Keypad.h → Reads input from 4x4 keypad.
 - Password.h → Stores and checks password.
- **Setup:**
 - Servo starts in locked position.
 - LCD shows welcome message, then prompts for password.
- **Working:**
 1. User enters **4-digit password** via keypad.
 2. Input is masked on LCD with *.
 3. If **correct password (1111)** → LCD shows success, servo rotates to unlock door, then auto re-locks.
 4. If **wrong password** → LCD displays error message and system resets.
 5. Pressing '**C**' key clears input and allows retry.

- **Functions:**

- showPrompt() → Prompts user to enter password.
- resetInput() → Clears previous input.
- unlockDoor() → Unlocks servo & shows success message.
- showError() → Displays wrong password message.

Workflow



How It Benefits and Useful in Real Life

- **Home Security:** Prevents unauthorized access using a secure, hard-coded or dynamically changeable password.
- **Office Access Control:** Employees can be given access with shared or individual PINs.
- **Remote Areas:** No need for internet; it can function locally and securely.
- **Elderly Friendly:** Easier than carrying keys which can be lost or forgotten.
- **Low Cost:** Affordable solution using easily available components.

Benefits

- **Cost-effective** compared to biometric or RFID systems.
- **Easy to use** and modify passwords if needed.
- **Compact and portable** system.
- **No keys required**, eliminating key-related issues.
- **Customizable:** Can integrate with GSM or Wi-Fi modules for alerts or remote access.

Applications of Unlocking Doors Using Sensors

1. Home Automation / Smart Homes

- **Sensors Used:** RFID, fingerprint, face recognition, or smartphone proximity (Bluetooth/NFC).
- **Purpose:** Provide secure, keyless entry for homeowners. Can be integrated with home security systems.

2. Office Buildings & Secure Facilities

- **Sensors Used:** Biometric (fingerprint, retina, face), RFID badge, keypad, or smartphone.
- **Purpose:** Restrict access to employees, maintain logs of entry/exit, enhance workplace security.

3. Hotels & Hospitality

- **Sensors Used:** RFID card readers, NFC via smartphones.
- **Purpose:** Replace traditional keys with keycards or mobile check-in apps.

4. Automobile Door Locks

- **Sensors Used:** Proximity sensors, key fobs (RFID/Bluetooth), fingerprint scanners (in newer models).
- **Purpose:** Allow contactless unlocking of car doors when authorized users approach.

5. Hospitals & Labs

- **Sensors Used:** Biometric scanners, RFID cards.

- **Purpose:** Control access to restricted areas (e.g., operating rooms, medicine storage).

6. Industrial or Warehouse Security

- **Sensors Used:** Facial recognition, fingerprint, RFID.
- **Purpose:** Prevent unauthorized access to machinery, hazardous zones, or inventory.

How It Overcomes Old Problems

Traditional Lock System	IoT Keypad Lock System
Can be picked	No physical keyhole
Keys can be lost or duplicated	Password-based access
No log of entry	Can be enhanced with logs
Difficult to manage access	Password can be changed anytime
Manual operation	Automatic with sensors

How It's Secured

- Passwords can be regularly updated or randomized.
- Limited attempts can be allowed before triggering a lockout or buzzer alert.
- Password input is hidden (e.g., shown as * * * *).
- Can be extended with features like:
 - GSM alert to owner on failed attempts.
 - Lockout period after multiple wrong tries.
 - Dual-factor authentication (Keypad + RFID/Bluetooth).

References

- Arduino Documentation: <https://www.arduino.cc>
- Keypad Arduino Library: <https://playground.arduino.cc/Code/Keypad/>
- Servo Motor Control: <https://www.arduino.cc/en/reference/servo>
- IoT Tutorials (Electronics Hub, CircuitDigest, etc.)

Picture of Final Model

